ISAM 102 Scotty Order of construction (2026 @ Rice)

Phase 1

1) Insert Bearings in Acrylic [L1, H5, H6, T8 (shared)]
    a) Flanged ball bearing flange on front face as marked on acrylic
    b) Flanged bushing with flange on front face as marked on acrylic
    c) Use mallet to hammer in bushing if needed
2) Bolt servo to acrylic [E1, 4 x H9, 4 x H24, 8 x H25, H20, 5 x T1, T4, T18]
    a) Insert servo into acrylic
    b) Insert M3 x 15mm bolt from print side
    c) On far side of bolt slide on washer
    d) On far side of bolt screw on locknut
    e) Repeat b, c, d 3 more times to complete motor mount
    f) Add ziptie at bottom for cable management
    g) Use diagonal flush cutters to cut off long end of zip tie

Phase 2

3) Chassis walls [L3, L7, L2, H9, H11]
    a) Bang in the scotty dog body (using a mallet)
    b) Snapfit the arch / captive nut slot
    c) Set acrylic assembly onto base
    d) Captive nut affix the arch (using M3 x 15mm)
4) Shoulder wall and captive nut mounting [L13, 4 x H9, 4 x H11]

Phase 3

5) Assemble 44-tooth gear and mounts [L6, D1, D7, 5 x H10, 5 x H11, H4]
    a) Mount clamping hubs on front, and non-clamping on back of gear
    b) Use M3 x 20mm nuts and bolts to make a sandwich of the gear
    c) Insert steel (rear leg) shaft, sticking ~¼" out of the hub
    d) Clamp the shaft onto the hub using an M3 nut and M3 x 20mm bolt
6) Insert rear leg shaft [D3, H7, H9, H11]
    a) Insert from acrylic side
    b) On scotty side, assemble with ¼" washer and tail rotor
    c) On scotty side, shaft may not extend beyond tail rotor
    d) Use M3 x 10mm bolt and nut to clamp tail rotor onto shaft
7) Build front-leg stack [D1, D6, L9, L14, H6, H10, H11, 4 x H26, H21]
    a) Screw clamping hub onto a cam of your choosing using M3 x 6mm screws (no nuts needed)
    b) Mount 3D printed gear onto wooden shaft using an M3 x 10mm screw through brass insert

c) Push flanged bushing into scotty side front leg hole
d) Insert shaft from 3D printed gear through bushing, so it's protruding on the scotty side
e) Wrap the exposed end of the shaft in 2–3 layers of masking tape to make it thicker
f) Mount cam assembly and tighten clamping hub to affix using an M3 x 20mm bolt

Phase 4

8) Screw on motor-mounted 12-tooth gear [L4, T3]
    a) Remove screw from servo
    b) Press-fit horn (attached to gear) onto motor
    c) Use screw from motor (that you just removed) to keep gear in place
9) Assemble tail [H3, H14, H28, D2, D9, D10, T7 and T12 (shared)]
    a) Cut cardboard
    b) Hot glue to skinny end of zip tie
    c) Use brass fastener to affix zip tie to end of wagger shaft, through the zip tie body, and flare far side of fastener tails to keep them in place
    d) Insert tail constraint ring

Phase 5

10) Connect head using binding barrel [H1, H2, L8, borrow 2nd T3 from neighbor]
11) Prepare battery pack [E8, 2 x H16, T2]
    a) Screw battery pack onto frame using small brass Phillips screws
12) Install potentiometer [E12]
    a) Take washer and nut off of it
    b) Orient the part correctly and slide it through the clearance hole
    c) Slide on the washer, screw on the nut (can use pliers to tighten)
13) Install rocker power switch [E14]
    a) Press fit the switch into the rectangular hole

Phase 6

14) Assemble breadboard circuit [E2, E3, E5, E6, E7, E13, E18, E21]
    a) Place Raspberry Pi Pico on breadboard, pin 1 of Pico into breadboard b1
    b) Use black jumper to connect pin 3 (ground) to the left side ground rail
    c) Use red jumper to connect pin 39 (Vsys) to right side power rail
    d) Insert long male–male header pins (3 in a row) into board, rows 23, 24, 25
    e) Use white jumper to jump from row 25, servo control pin, to Raspberry Pi pin 20
    f) Use red jumper from center servo pin (row 24) to power rail
    g) Use black jumper from leftmost servo pin (row 23) to ground rail
    h) Insert LTH 1550-01 IR proximity sensor into breadboard, clear side legs into 29f and 30f, black side legs into 29e and 30e
    i) Insert 220Ω resistor from 29j into power rail
    j) Insert 5.6kΩ resistor from 29a into ground rail
    k) Use black jumper from 30g to ground rail
    l) Use red jumper from 30d to power rail

m) Use blue jumper from 29d to Raspberry Pi pin 31 (10j)
n) Insert 10Ω resistor from 19h to 21f

15) Move breadboard into place and make final connections
a) Remove tape guard and stick breadboard onto base
b) Connect potentiometer black leg to ground rail
c) Connect potentiometer red leg to power rail
d) Connect potentiometer green (signal) leg to Raspberry Pi pin 32 (9i)
e) Plug servo motor in to male pins (23c, 24c, 25c)
f) Plug battery pack black wire (ground) into breadboard ground rail

16) Attach power switch [E15, E20]
a) Push spade connector onto top terminal of power switch
b) Feed short red wire from spade connector into Wago connector, clip shut
c) Feed red wire from battery pack into Wago connector, clip shut
d) Attach long switch red wire to board power rail

17) Pet the dog [5 x D8, 3 x E9]
a) Press-fit knob onto potentiometer shaft (use a bit of tape to enlarge the shaft if needed)
b) Insert 3 AA batteries into holder
c) Turn power switch to “I”
d) See if you get motion and a blinking power LED (they should vary based on the min. speed potentiometer position as well as detected proximity)

18) Install LED [E10, E11]
a) Use heat gun to shrink orange heat shrink around LED wires
b) Push fresnel lens into hole in dog head from the front side
c) Insert LED into lens from back side
d) Route wires under dog body, connect to 19j and 20j

19) Do zip-tie wire tidying [H23, 3 x H20]
a) Use zip ties to affix LED wires to acrylic leaving slack at the top so head can rotate
b) Mount zip tie sticky base near scotty front leg
c) Use zip tie to hold wires to base
d) Use flush diagonal cutter to cut off long ends of zip ties